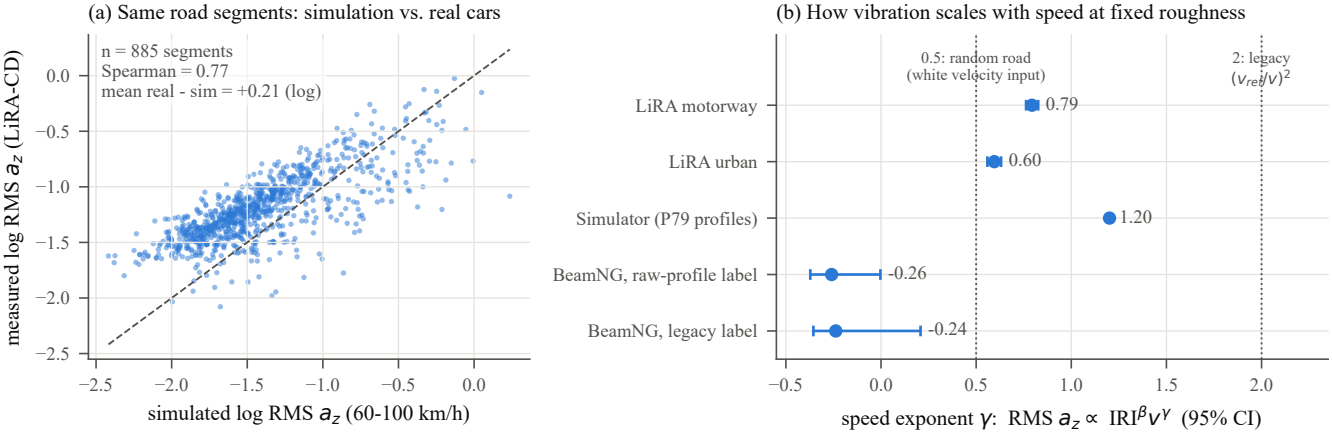

(a) Same road segments: simulation vs. real cars
n = 885 segments
Spearman = 0.77
mean real - sim = +0.21 (log)
measured log RMS a_z (LiRA-CD)
simulated log RMS a_z (60-100 km/h)
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
(b) How vibration scales with speed at fixed roughness
0.5: random road
(white velocity input)
2: legacy
$(v_{ref}/v)^2$
LiRA motorway
LiRA urban
Simulator (P79 profiles)
BeamNG, raw-profile label
BeamNG, legacy label
0.79
0.60
1.20
-0.26
-0.24
−0.5
0.0
0.5
1.0
1.5
2.0
speed exponent γ: RMS $a_z \propto$ IRI$^{\beta} v^{\gamma}$ (95% CI)